Programming Project Proposal

Project Title: Adversarial Attacks and Defenses on Cloud-Based Medical Diagnostics with LLM Advisory

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Project Description:
We propose to build a cloud-hosted image classification system that can diagnose medical conditions using a publicly available dataset, such as Chest X-ray images. Since training and running deep learning models on large datasets requires significant computing power, many organizations rely on third-party cloud services to reduce costs. While this approach is efficient, it also introduces security risks because sensitive medical data and machine learning models operate in an environment that cannot always be fully trusted. As a result, the system may be vulnerable to insider threats within the cloud provider or attacks that occur while data is being transmitted.

Our project will begin by training a neural network on clean Chest X-ray images so that it can accurately classify different medical conditions. After deploying the trained model in a simulated cloud environment, we will evaluate its security by performing a test-time adversarial attack. In this attack, an adversary subtly modifies the input X-ray by adding carefully crafted, nearly invisible mathematical noise. Although these changes are imperceptible to humans, they can mislead the neural network into making an incorrect diagnosis, demonstrating how attackers could potentially exploit AI systems for purposes such as insurance fraud or other malicious activities.

To improve the system's resilience, we will develop a defense framework that incorporates a Large Language Model (LLM) as an intelligent security advisor. Whenever suspicious behavior or a potential adversarial attack is detected, the system will provide the LLM with information about the attack. Based on this context, the LLM will recommend the most appropriate defense strategy. These recommendations may include introducing randomized defenses to reduce the effectiveness of transferable adversarial attacks or deploying an Adversarial Support Vector Machine (ADV-SVM) to strengthen the model's decision boundaries. The project will be implemented in Python using open-source deep learning libraries, an LLM API, and Anthropic's Claude Code to assist with software development and system integration.